

Compute Before Retransmit: Reliability-Anchored Flow Matching for DeepJSCC HARQ

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Abstract—Can a receiver improve its current estimate enough to avoid another physical transmission? We study this question for image DeepJSCC with two-round, quality-based hybrid automatic repeat request (HARQ). FlowHARQ repairs selected latent tokens through conditional flow matching, preserves unselected tokens, and applies a separately calibrated quality decision. A fixed codec and common maximum-ratio-combining fallback isolate the effect of receiver computation. On DIV2K and Kodak24, an existing ten-run extension reduces NACK probability by 1.319 and 1.453 percentage points, respectively, within predefined mean PSNR and LPIPS tolerances. A paired decision decomposition shows that quality improvement on already-accepted images offsets degradation on saved rounds; the aggregate is not a per-image improvement. Charging token-scale metadata and reliable feedback reduces payload savings of 0.758% and 0.846% to 0.353% and 0.391%. Substantial false acceptance and final outage remain. A paired published-SwinJSCC transfer test also exposes a large codec-quality gap. The evidence supports a small within-codec tradeoff, not reliable target-quality delivery or a competitive end-to-end HARQ system.

Index Terms—DeepJSCC, HARQ, flow matching, semantic communication, receiver processing.

I. INTRODUCTION

Deep joint source–channel coding (DeepJSCC) produces a reconstruction even when channel conditions do not support high image quality [1]. A receiver must therefore decide whether to accept that reconstruction or request additional symbols. This decision has two distinct failure modes: requesting a transmission that local processing could avoid, and accepting an image whose quality falls below the service target. Reducing the first without measuring the second can make an overconfident receiver appear efficient.

We investigate a receiver action between equalization and ACK/NACK: refine the received latent representation, then reassess whether a physical round is needed. The action adds computation but no channel observation. It is useful only if its change in the final decision compensates for its processing cost without an unacceptable quality loss. Image restoration quality alone cannot establish this communication benefit.

Position relative to prior work. DeepJSCC-f uses channel-output feedback to encode further information [2], while SAFG-HARQ changes retransmission granularity [3]. These approaches modify physical information delivery. CDDM instead uses diffusion for channel denoising [4]; FlowSem and realization-coupled flow matching study generative refinement in communication [5], [6], and generative semantic HARQ has been investigated for text latents [7]. In computer vision, FlowIE studies conditional rectified flow for efficient

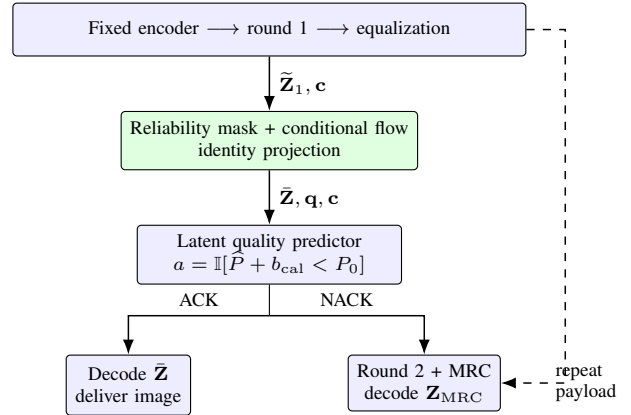


Fig. 1. Receiver computation precedes the physical retransmission decision. Both policies use the same first observation and identical MRC fallback. The dashed path carries the original payload, not newly generated information.

enhancement [8]; ResFlow models restoration through an augmented degradation flow [9]. These works motivate iterative refinement, but their restoration results do not establish safe early acceptance on a wireless link. Our question concerns the subsequent communication decision, not priority for applying flow matching.

FlowHARQ constrains repair with a reliability mask, then decides on the repaired state. On NACK, it uses the same two-observation MRC fallback as the no-repair policy. When both retransmit, their delivered images are identical.

The paper makes three contributions. First, it specifies this receiver action with explicit forward symbols, side information, and binary feedback. Second, it separates changes in accepted-image quality from changes due to saved or additional rounds through a paired decision decomposition. Third, it evaluates the resulting tradeoff using frozen experiments, a finite-precision audit, and charged signaling costs. The benefit is modest and does not establish reliable 24-dB delivery. We report that boundary alongside the saving rather than equating an average-quality tolerance with a reliability guarantee.

II. SYSTEM MODEL AND EVALUATION OBJECTIVE

A. Two-Round Image Link

For an RGB image $\mathbf{x} \in [0, 1]^{H \times W \times 3}$, a fixed encoder produces $\mathbf{Z} = E_\theta(\mathbf{x}) \in \mathbb{R}^{N \times d}$. Pairing real features forms $d/2$ complex symbols $\mathbf{z}_i^c$ per token. The transmitted token is $\mathbf{s}_i = \mathbf{z}_i^c / \alpha_i$, where $\alpha_i = (2\|\mathbf{z}_i^c\|^2/d + \epsilon)^{1/2}$ and $\epsilon = 10^{-8}$. For $N = 256$, $d = 32$, and $H = W = 256$, each round

carries $B = Nd/2 = 4096$ complex payload symbols, giving a payload channel-bandwidth ratio of $1/48$.

Round $r \in \{1, 2\}$ yields $\mathbf{y}_r = h_r \mathbf{s} + \mathbf{n}_r$, with $\mathbf{n}_r \sim \mathcal{CN}(0, 10^{-\gamma/10} \mathbf{I})$ at nominal SNR γ dB. One flat-fading coefficient applies to the image. Equalization uses perfect instantaneous CSI and restores the token scales:

$$\tilde{\mathbf{z}}_{r,i}^c = \alpha_i \frac{h_r^* \mathbf{y}_{r,i}}{|h_r|^2 + \epsilon}. \quad (1)$$

The original experiment uses exact scales. Our signaling audit charges $Q_{16}(\alpha_i)$, an IEEE binary16 value, transmitted once and reused if the same payload is retransmitted; the finite-precision check replaces α_i by $Q_{16}(\alpha_i)$ in the receiver restoration only.

The two rounds follow a correlated Rayleigh law,

$$h_2 = \rho_\Delta h_1 + \sqrt{1 - \rho_\Delta^2} e, \quad \rho_\Delta = J_0(2\pi v f_c \Delta / c), \quad (2)$$

where $h_1, e \sim \mathcal{CN}(0, 1)$ independently, v is speed in m/s, $f_c = 5.9$ GHz, and $\Delta = 1$ ms. Speed changes inter-round correlation, not the marginal first-round fading law. Context $\mathbf{c}$ comprises SNR, speed, CSI age and its Jakes correlation, instantaneous gain, and effective SNR. CSI age is a conditioning variable, not simulated estimation error.

A NACK requests the same normalized payload and produces

$$\mathbf{Z}_M = \frac{|h_1|^2 \tilde{\mathbf{Z}}_1 + |h_2|^2 \tilde{\mathbf{Z}}_2}{|h_1|^2 + |h_2|^2}, \quad (3)$$

with numerical denominator protection in code. The decoder D_θ delivers an accepted first-round state or $D_\theta(\mathbf{Z}_M)$. No third round is allowed. ACK/NACK is reliable; this is image-quality HARQ, not CRC-based error detection. The target $P_0 = 24$ dB need not be attained after the last allowed round.

B. Charged Channel Uses and Quality Constraints

Let $a = 1$ denote a NACK. With metadata and feedback efficiencies η_s, η_f measured in information bits per complex channel use, define

$$C(a) = B(1 + a) + \left\lceil \frac{16N}{\eta_s} \right\rceil + (1 + a) \left\lceil \frac{1}{\eta_f} \right\rceil. \quad (4)$$

The last term charges the initial ACK/NACK and a final ACK when a second round occurs. Metadata is reused rather than retransmitted. These are charged uses under ideal reliable signaling, not a complete air-interface budget: pilots, framing, feedback delay, and control decoding errors are excluded.

Relative to a fixed adaptive-HARQ policy A , we evaluate receiver policy F by the design objective

$$\min_F \mathbb{E}[C(a_F)] \\ \text{s.t. } \mathbb{E}[P_A - P_F] \leq 0.02 \text{ dB}, \quad \mathbb{E}[L_F - L_A] \leq 0.003, \quad (5)$$

where P and L are delivered-image PSNR and LPIPS [10]. This is an evaluation criterion, not a claim that our staged training solves a constrained optimum. Neither average tolerance supplies an image-wise outage guarantee. Writing

$p_j = \Pr(a_j = 1)$ and $b_f = \lceil 1/\eta_f \rceil$, the exact accounting identity is

$$\mathbb{E}[C(a_A) - C(a_F)] = (B + b_f)(p_A - p_F). \quad (6)$$

Fixed metadata cancels in the absolute difference but dilutes relative savings through the denominator. Thus $p_A - p_F$, $(p_A - p_F)/p_A$, and $(p_A - p_F)/(1 + p_A)$ represent different quantities: an absolute NACK change, a relative reduction in requests, and a relative payload saving.

C. What a Saved Round Does and Does Not Establish

Let P_D, P_R, P_M denote the PSNR of direct, repaired, and MRC images for the same source and physical observations. With $I_{uv} = \mathbb{I}[a_A = u, a_F = v]$, the delivered-quality difference obeys the sample-wise identity

$$P_F - P_A = I_{00}(P_R - P_D) + I_{10}(P_R - P_M) + I_{01}(P_M - P_D). \quad (7)$$

This follows by enumerating the four decision pairs; the I_{11} term is zero because the fallback is common. The same identity holds for any scalar reconstruction metric, including MSE and LPIPS, with their respective signs. An overall PSNR gain can therefore coexist with a loss on the saved-round subset. We evaluate both the net change and its components.

For reliability, define ACK coverage $\kappa_j = \Pr(a_j = 0)$, joint false ACK $u_j = \Pr(a_j = 0, P_j < P_0)$, and conditional false ACK $r_j = u_j/\kappa_j$ when $\kappa_j > 0$. Final outage is $o_j = \Pr(P_j < P_0)$, including delivery after MRC. Neither (5) nor the cost identity bounds r_j or o_j . When $\kappa_j = 0$, conditional false ACK is undefined, not evidence of a useful zero-risk service. These quantities distinguish economical decisions from reliable delivery.

III. RELIABILITY-ANCHORED FLOWHARQ

A. Repair Only the Selected Latent Coordinates

A reliability head predicts $q_i = g_q(\tilde{\mathbf{z}}_{1,i}, \mathbf{c})$, and inference selects $m_i = \mathbb{I}[q_i \geq \tau_r]$. During training only, an oracle mask m_i^* selects the largest $\lceil 0.35N \rceil$ clean-to-received token errors and supervises a balanced binary cross-entropy loss. Neither the clean image nor this oracle mask is available to the deployed receiver.

Following conditional flow matching [11], let $\mathbf{u}_i = \mathbf{z}_i - \tilde{\mathbf{z}}_{1,i}$ and sample $t \sim \mathcal{U}[0, 1]$ on the path $\mathbf{z}_i(t) = \tilde{\mathbf{z}}_{1,i} + tm_i^* \mathbf{u}_i$. The velocity network sees this state, time, receiver context, the received tokens, and a pooled anchor from unselected tokens. To avoid domination by large zero-forcing residuals in deep fades, its regression loss is

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_t \frac{\sum_i m_i^* d^{-1} \sum_j \rho_{0.1}((v_{\psi,ij} - u_{ij})/s_i)}{\max(1, \sum_i m_i^*)}, \\ s_i = \max(1, \|\mathbf{u}_i\|/\sqrt{d}), \quad (8)$$

where $\rho_{0.1}$ is smooth- L_1 and s_i is detached. An image L_1 loss on decoded repair aligns transport with the fixed decoder.

That branch uses the predicted hard mask with a straight-through gradient; it does not use the oracle mask from velocity supervision.

At inference, $K = 4$ explicit updates start from $\mathbf{Z}^{(0)} = \tilde{\mathbf{Z}}_1$:

$$\mathbf{Z}^{(k+1)} = \mathbf{Z}^{(k)} + \frac{\mathbf{m}}{K} \odot v_\psi(\mathbf{Z}^{(k)}, (k + \frac{1}{2})/K; \tilde{\mathbf{Z}}_1, \mathbf{m}, \mathbf{c}). \quad (9)$$

Mid-interval time inputs do not make this a second-order mid-point solver. The output projection $\tilde{\mathbf{z}}_i = (1 - m_i)\tilde{\mathbf{z}}_{1,i} + m_i\mathbf{z}_i^{(K)}$ preserves every unselected latent token exactly. It does not preserve spatial image regions, since decoding mixes tokens. Likewise, the dense Transformer still processes all tokens; masking constrains content changes rather than delivering sparse-computation savings.

B. Decide on the Repaired State, Fall Back to Physical Evidence

A quality head estimates log-MSE from the latent, reliability scores, and context. Its converted PSNR prediction $\hat{P}$ is trained on direct and repaired states using log-MSE regression and a boundary loss

$$\mathcal{L}_q^{\text{boundary}} = \text{BCEWithLogits}((P_0 - \hat{P})/T_q, \mathbb{I}[P < P_0]). \quad (10)$$

The true quality on the right is training supervision only. A separate decision-training stage freezes the codec, reliability head, and velocity field. Scalar biases b_A, b_F are subsequently fitted on calibration images, giving $a_A = \mathbb{I}[\hat{P}(\tilde{\mathbf{Z}}_1) + b_A < P_0]$ and $a_F = \mathbb{I}[\hat{P}(\tilde{\mathbf{Z}}) + b_F < P_0]$. On ACK, FlowHARQ decodes $\tilde{\mathbf{Z}}$; on NACK it decodes (3). The baseline uses the direct state on ACK. Separate calibration is necessary because repair changes the predictor’s input distribution; calibration does not, by itself, certify a false-ACK bound.

IV. EXPERIMENTAL RESULTS

A. Protocol and Comparisons

We test request savings, their decision-level mechanism, and their reliability and signaling costs. The fixed Swin-based codec has 21.83M parameters, while the reliability, velocity, and quality heads add 0.479M. The velocity field has two Transformer blocks, width 128, and four attention heads. Receiver training uses DIV2K [12] random 256×256 crops for 40 epochs, followed by ten quality-head epochs. AdamW uses batch size 2, learning rate 2×10^{-4} for repair and 10^{-4} for the quality-only stage, weight decay 10^{-4} , and gradient clipping at 5. The boundary temperature is 1 dB, with equal regression and classification weights. Training samples SNR uniformly over 0–15 dB, speed over 0–120 km/h, and CSI age over 0–5 ms. Calibration fixes $K = 4$, $\tau_r = 0.8$, $P_0 = 24$ dB, and the quality biases using the first 20 DIV2K validation images. The remaining 80 images are held out; Kodak24 receives no recalibration. Test images are center-cropped to 256×256 RGB. Tests span SNRs $\{0, 3, 6, 9, 12, 15\}$ dB and speeds $\{30, 60, 90, 120\}$ km/h, with CSI age fixed at 2 ms.

The original confirmation comprises three trained receivers and three channel seeds per receiver. A previously locked extension increases the receiver count to ten; it is reported

TABLE I
PAIRED FLOWHARQ MINUS ADAPTIVE HARQ. POSITIVE SAVING MEANS FEWER NACKS; POSITIVE ΔLPIPS MEANS WORSE PERCEPTUAL DISTANCE. ORIGINAL AND EXTENDED COHORTS ARE SEPARATED. PSNR IS IN DB.

Dataset	Runs	Saving (pp) [95% CI]	ΔPSNR	ΔLPIPS
DIV2K	3	1.296 [0.023, 2.570]	+0.0206	+0.00132
DIV2K	10	1.319 [1.102, 1.537]	+0.0144	+0.00143
Kodak24	3	2.025 [0.708, 3.343]	+0.0284	+0.00128
Kodak24	10	1.453 [1.081, 1.825]	+0.0253	+0.00131

separately, not relabeled as the original experiment. No additional training or seed sweep is performed for this paper revision. Each policy is evaluated on paired images and physical channel observations. Confidence intervals in Table I use paired per-training-run differences with a Student- t interval, averaging channel draws within each run. Repeated channel draws are not counted as independent training runs.

Controlled comparisons comprise direct one-round decoding, always-two-round MRC, repair without HARQ, adaptive HARQ without repair, and FlowHARQ. These isolate the receiver decision with identical codec and channel resources. They are not independent published architectures. Published author-model references are treated separately below because feedback, training, and channel assumptions differ.

B. Does Receiver Repair Reduce Physical Requests?

Table I separates the original result from the ten-run extension. In the latter, DIV2K NACK probability changes from 73.970% to 72.651%, and Kodak24 from 71.701% to 70.249%. Mean delivered PSNR changes by +0.0144 and +0.0253 dB, respectively. LPIPS changes by +0.00143 and +0.00131: both are within the 0.003 tolerance, but neither is a perceptual improvement. Fig. 2 gives the SNR dependence rather than presenting the aggregate as a uniform gain.

C. Where Does the Gain Come From?

To locate the effect, partition paired samples by (a_A, a_F) . The identity $p_A - p_F = \Pr(1, 0) - \Pr(0, 1)$ separates saved from newly requested rounds. On DIV2K, 1.4740% of samples save a round and 0.1545% request an extra one, leaving the net 1.3194-point reduction. Both receivers accept 25.8750% and both request retransmission on 72.4965%. In the last group the delivered images are identical, because both use the same MRC observation.

Equation (7) explains the quality result. DIV2K samples accepted by both policies contribute +0.02719 dB to the overall mean PSNR difference; saved-round samples contribute −0.01332 dB, and extra-round samples +0.00054 dB. Thus the net PSNR gain does not mean that skipping a physical observation improves its reconstruction. It combines a quality benefit on already-accepted images with a small quality sacrifice on saved rounds. Kodak24 has the same structure: 1.5856% saved rounds versus 0.1331% extra rounds. This is an average resource–quality tradeoff, not per-image dominance.

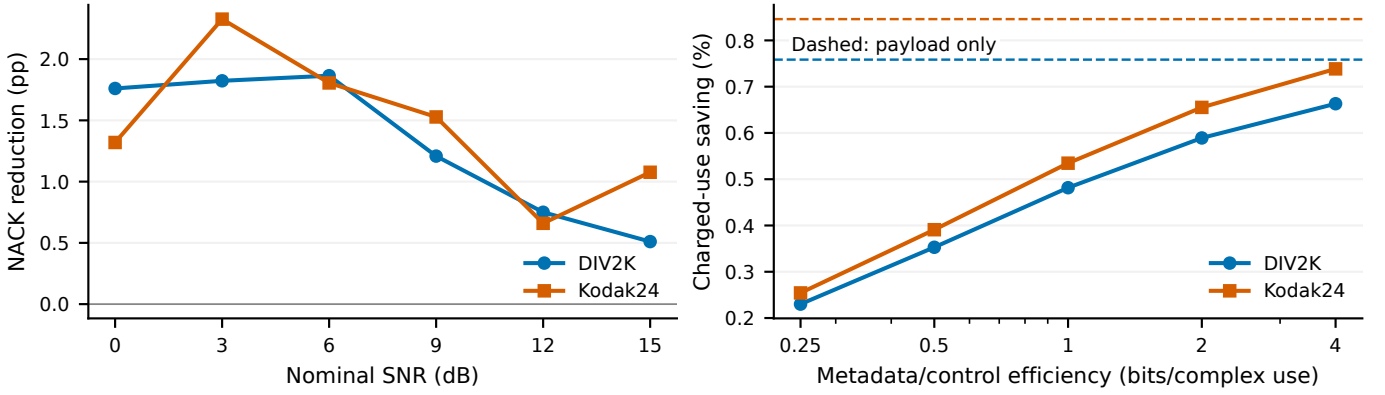


Fig. 2. Frozen ten-run extension, uniformly averaged over images, four speeds, and three channel draws per trained receiver. Left: paired NACK reduction by SNR. Right: relative saving after charging binary16 token scales once and reliable control at $\eta_s = \eta_f$; dashed lines show payload-only savings. The right panel re-accounts archived decisions and is not a new quantized-model performance experiment. All efficiency points are reported.

D. Does the Saving Imply Reliable Delivery?

In the ten-run extension, conditional false ACK changes from 13.51% to 14.03% on DIV2K and from 50.88% to 50.57% on Kodak24. FlowHARQ’s final outage is 70.89% and 82.92%, respectively. These figures include difficult low-SNR fades and show that 24 dB is not a reliably met service level. Lower average NACK cost within mean-quality tolerances is therefore not evidence of reliable early stopping.

A post-hoc frozen-receiver diagnostic tests whether a more conservative decision resolves this problem. For receiver 2030, calibration alone selects a 1-dB extra margin from $\{0, 0.25, 0.5, 1, 2, 4\}$ dB, requiring empirical conditional false ACK below 5% and at least 5% ACK coverage. Both policies select the same margin. Across the three archived channel streams, DIV2K conditional false ACK is 4.97% for adaptive HARQ and 3.86% for FlowHARQ, but Kodak24 reaches 31.89% and 30.32%. Thus a calibration safety margin does not transfer reliably; these empirical values are not risk certificates.

E. Does the Saving Survive Signaling Costs?

The ten-run NACK reductions correspond to 1.784% and 2.026% fewer second-round requests, but only 0.758% and 0.846% fewer payload symbols. With $\eta_s = \eta_f = 0.5$, 256 binary16 scales cost 8192 complex uses once per image, exceeding a single payload round. Equation (4) then gives 0.353% and 0.391% relative charged-use savings. This overhead is a material limitation of the current token-normalized codec. Compressing the scales or changing normalization could improve efficiency, but would change the implemented system and is not credited here.

The original archive records single-scale SSIM, not MS-SSIM. Table II reports a separate frozen-checkpoint audit on Artemis: one existing trained receiver, one channel stream per dataset, all 24 SNR–speed conditions, and paired ideal/binary16 scale restoration. CPU random streams differ from the archived CUDA streams; this is a precision check, not their exact replay.

TABLE II

FROZEN BINARY16 AUDIT UNDER CORRELATED RAYLEIGH FADING. ONE EXISTING RECEIVER AND ONE CHANNEL STREAM PER DATASET, WITH 1920 DIV2K AND 576 KODAK IMAGE–CHANNEL CASES PER METHOD. NO UNCERTAINTY ESTIMATE OR NEW TRAINING IS INFERRED FROM THESE ROWS.

Policy	NACK (%)	PSNR	MSE $\times 10^3$	MS-SSIM	LPIPS
<i>DIV2K</i>					
Direct	0.00	21.319	10.239	0.8000	0.5127
Full HARQ	100.00	21.985	8.587	0.8405	0.4827
Repair only	0.00	21.392	10.165	0.7972	0.5164
Adaptive	74.69	21.912	8.630	0.8379	0.4853
FlowHARQ	73.02	21.948	8.608	0.8371	0.4867
<i>Kodak24</i>					
Direct	0.00	21.470	8.554	0.7797	0.5575
Full HARQ	100.00	22.105	7.142	0.8259	0.5189
Repair only	0.00	21.580	8.409	0.7760	0.5626
Adaptive	74.31	22.033	7.191	0.8220	0.5238
FlowHARQ	71.70	22.070	7.165	0.8209	0.5257

Quantization changes 0 of 3840 paired adaptive/FlowHARQ decisions on DIV2K and 0 of 1152 on Kodak24. Across all five methods and both datasets, the largest per-image PSNR change is 0.0250 dB. PSNR and MSE alone do not establish perceptual benefit; the measured MS-SSIM and LPIPS must be considered separately. No threshold or weight is refitted using this audit, and it does not replace the multi-run confirmation.

F. What Do the Controls and Published Models Establish?

An existing parameter-matched control trained at flow time zero yields -0.0147 dB at its best tested calibration threshold, versus $+0.0714$ dB for the earlier four-step recipe. Different time supervision and threshold selection prevent isolating integration depth. This calibration comparison supports retaining the measured recipe, not claiming that flow matching is necessary or that a held-out one-step alternative has been ruled out.

Table III provides a new paired channel-transfer diagnostic using published SwinJSCC C32 weights [13]. Both codecs receive the same two fading gains and noise vectors, with 4096 complex payload symbols per round. The six-SNR/four-speed

TABLE III

PAIRED RAYLEIGH TRANSFER DIAGNOSTIC, ONE DRAW PER IMAGE/CONDITION. AUTHOR SWINJSCC USES ONE OR TWO ROUNDS (1R/2R); 2R IS OUR CHASE WRAPPER. USES INCLUDE BINARY16 SCALES AND RELIABLE CONTROL. TRAINING RECIPES DIFFER.

Data	Receiver	PSNR	MSE $\times 10^3$	MS-SSIM	LPIPS	Uses
DIV	Swin 1R	23.07	25.345	0.7730	0.3463	4130
DIV	Swin 2R	26.14	9.807	0.8852	0.2641	8228
DIV	FlowHARQ	21.94	8.653	0.8361	0.4873	15197
Kodak	Swin 1R	23.02	21.348	0.7506	0.3758	4130
Kodak	Swin 2R	26.34	5.988	0.8851	0.2771	8228
Kodak	FlowHARQ	22.05	7.193	0.8204	0.5258	15171

grid and image crops are unchanged. We use one existing FlowHARQ receiver and one channel draw per case on an RTX 4090, without retraining or recalibration. SwinJSCC retains its global packet scale, whereas FlowHARQ retains 256 token scales; both are binary16 and charged. The two-round SwinJSCC variant uses our Chase/MRC wrapper, not a published HARQ architecture. Different training recipes and normalization remain explicit.

The comparison exposes a codec-level limitation. On Kodak24, two-round SwinJSCC reaches 26.34 dB versus FlowHARQ's 22.05 dB with lower MSE and LPIPS, higher five-scale MS-SSIM [14], and 8228 versus 15171 charged uses. On DIV2K, its mean PSNR is also higher, but mean MSE is worse: averaging logarithmic PSNR can conceal large errors in deep fades. Thus neither a within-codec gain nor mean PSNR alone establishes system superiority. Separate archived AWGN evaluations of NTSCC [15], CDDM, and DeepJSCC-f retain their distinct rate and feedback assumptions; they are not substituted for a matched published HARQ comparison.

G. Scope and Limitations

Imperfect CSI, corrupted metadata, and shifted image distributions remain deployment risks. The original RTX PRO 6000 microbenchmark gives 3.11 ms for decoding and 5.00 ms for repair plus decoding, excluding the quality head and radio. For additional processing time δT and round time T_R , the simplified break-even condition is $\delta T < (p_A - p_F)T_R$; neither end-to-end latency nor energy savings are measured. Selective retransmission and joint rate-aware training are outside the validated claim.

V. CONCLUSION

FlowHARQ isolates receiver refinement before ACK/NACK through a common physical fallback. Frozen experiments support small within-codec savings under mean-quality tolerances, but decision decomposition, acceptance errors, and metadata costs limit their interpretation. The published-codec comparison exposes a larger system gap. Validation on a strong pretrained codec at matched acceptance risk, followed by a faithful external HARQ comparison, remains necessary.

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